COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
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ACTIVITY NO. 10 **Install, Configure and Manage Enterprise Log Monitoring via Ansible**

1. PROCEDURE OUTPUT

1. Create a playbook that:
a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.
5. Make sure to create a new repository in GitHub for this activity.

OUTPUT:

Step 1: **Directory and Inventory Structure**

```
leroy1@leroyque:~/CPE212_Activity10$ tree
.
├── ansible.cfg
├── inventory.ini
├── roles
│   ├── elasticsearch
│   │   └── tasks
│   │       └── main.yml
│   ├── kibana
│   │   └── tasks
│   │       └── main.yml
│   └── logstash
│       └── tasks
│           └── main.yml
└── site.yml

8 directories, 6 files
```

```
leroy1@leroyque:~/CPE212_Activity10$ cat inventory.ini
[elasticsearch]
ubuntu_elastic ansible_host=192.168.56.104 ansible_user=leroy1

[kibana]
ubuntu_kibana ansible_host=192.168.56.106 ansible_user=leroy1

[logstash]
centos_logstash ansible_host=192.168.56.107 ansible_user=sysadleroy1
```

Step 2: **Playbook Execution**
site.yml

```
leroy1@leroyque:~/CPE212_Activity10$ cat site.yml
---
- name: Install and configure Elasticsearch
  hosts: elasticsearch
  become: true
  roles:
    - elasticsearch

  tasks:
    - name: Wait for Elasticsearch to become available
      wait_for:
        host: 192.168.56.104
        port: 9200
        delay: 5
        timeout: 60

- name: Install and configure Kibana
  hosts: kibana
  become: true
  roles:
    - kibana

  tasks:
    - name: Wait for Kibana to become available
      wait_for:
        host: 192.168.56.106
        port: 5601
        delay: 5
        timeout: 60

- name: Install and configure Logstash
  hosts: logstash
  become: true
  roles:
    - logstash
```

Analysis: This playbook site.yml manages the execution of roles in the specified host groups. The roles are to be executed sequentially with all privileges (become: true). Besides, in the playbook, the wait_for module is employed in case of TCP ports 9200 and 5601 to check whether required services are running, so that the next tasks would be executed. Thus, race conditions would not occur in the process of the automated deployment of the services.

```
TASK [logstash : Install Logstash] *********************************************************************************************************
ok: [centos_logstash]

TASK [logstash : Enable and start Logstash] ************************************************************************************************
ok: [centos_logstash]

PLAY RECAP *********************************************************************************************************************************
centos_logstash            : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu_elastic             : ok=9    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu_kibana              : ok=9    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Analysis: The playbook output shows that all tasks finished with zero failure (failed=0). Which means, it is proven that everything was installed correctly on both Ubuntu and CentOS.

Step 3: **Elastic Verification (Ubuntu Host 1)**
roles/elasticsearch/tasks.main.yml

```
leroy1@leroyque:~/CPE212_Activity10$ cat roles/elasticsearch/tasks/main.yml
---
- name: Install required dependencies
  apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
      - curl
    state: present
    update_cache: true

- name: Download Elastic GPG key
  shell: |
    wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

- name: Add Elastic 8.x repository
  apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/8.x/apt stable main"
    filename: elastic-8.x
    state: present

- name: Install Elasticsearch
  apt:
    name: elasticsearch
    state: present
    update_cache: true
```

```
- name: Deploy clean Elasticsearch configuration
  copy:
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'
    content: |
      cluster.name: my-application
      node.name: node-1
      path.data: /var/lib/elasticsearch
      path.logs: /var/log/elasticsearch
      network.host: 0.0.0.0
      http.port: 9200
      discovery.type: single-node
      xpack.security.enabled: false
      xpack.security.enrollment.enabled: false
      xpack.security.http.ssl.enabled: false
      xpack.security.transport.ssl.enabled: false

- name: Reload systemd units
  systemd:
    daemon_reload: true

- name: Enable and start Elasticsearch
  systemd:
    name: elasticsearch
    enabled: true
    state: restarted
```

Analysis: This task imports the official key from Elastic and sets up the software repository for the 8.x version on Ubuntu. It configures the installation of Elasticsearch as a single node listening on 0.0.0.0 with the security configurations off during testing. It concludes by requesting systemd to reload the configuration and start Elasticsearch.

```
leroy1@leroyque:~/CPE212_Activity10$ curl http://192.168.56.104:9200
{
  "name" : "node-1",
  "cluster_name" : "my-application",
  "cluster_uuid" : "xnByc_zyTYiszOrOliT3nA",
  "version" : {
    "number" : "8.19.22",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "3b2a41103de35e0af4064d647974032fcc1bcde9",
    "build_date" : "2026-09-18T10:10:07.018982263Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.2",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

Analysis: For running this curl command, it will return a JSON response in which this includes the name of the node alongside the version 8.19.22. This shows that Elasticsearch is up and running on port 9200.

Step 4: **Kibana / Elastic Dashboard Verification (Ubuntu Host 2)**
roles/kibana/tasks/main.yml

```
leroy1@leroyque:~/CPE212_Activity10$ cat roles/kibana/tasks/main.yml
---
- name: Install required packages
  apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
      - curl
    state: present
    update_cache: true

- name: Download Elastic GPG key
  shell: |
    wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

- name: Add Elastic 8.x repository
  apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/8.x/apt stable main
    filename: elastic-8.x
    state: present

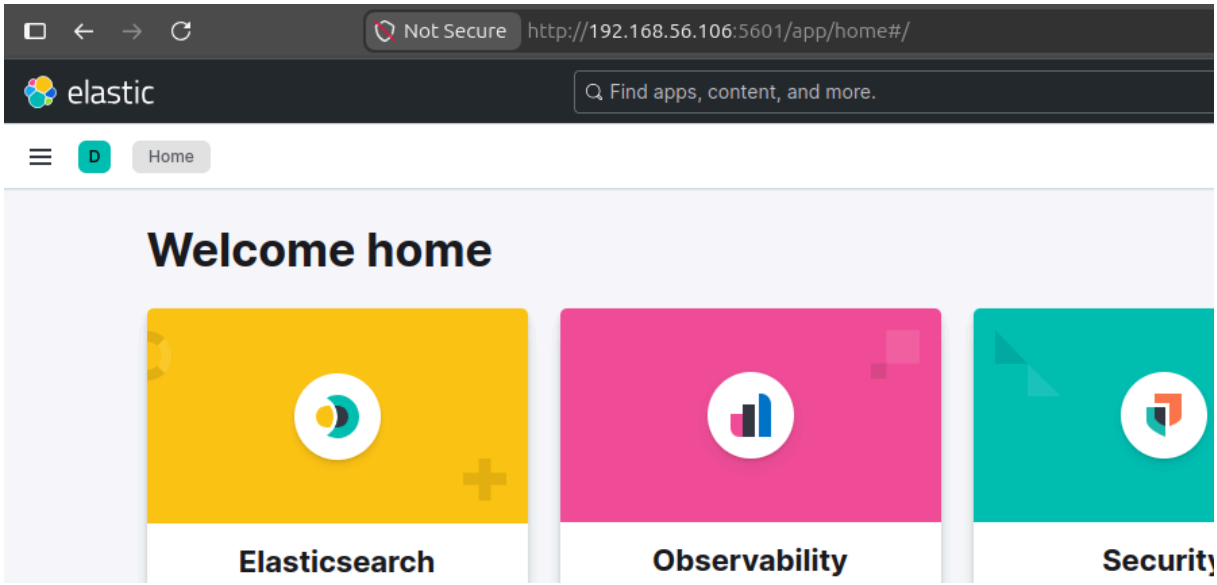
- name: Install Kibana
  apt:
    name: kibana
    state: present
    update_cache: true

- name: Configure Kibana network host
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?server.host:'
    line: 'server.host: "0.0.0.0"'
```

```
- name: Configure Kibana Elasticsearch host URL
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?elasticsearch.hosts:'
    line: 'elasticsearch.hosts: ["http://192.168.56.104:9200"]

- name: Enable and start Kibana
  systemd:
    name: kibana
    enabled: true
    state: restarted
    daemon_reload: true
```

Analysis: This task of installing Kibana and modifying its configuration is completed through the use of Ansible's lineinfile function. The connection to the server is made through the host IP of 0.0.0.0, which makes it accessible through its 5601 port while providing the link to http://192.168.56.104:9200 where it can be linked to the Elasticsearch VM. Kibana is then rebooted in order to ensure it functions automatically when the system starts.



Analysis: By typing the said url in the browser, the homepage of Elastic should be accessed. In which if it is successful, this just shows that Kibana has been installed and connected to Elasticsearch.

Step 5: Logstash Service Verification (CentOS Host)
roles/logstash/tasks/main.yml

```
leroy1@leroyque:~/CPE212_Activity10$ cat roles/logstash/tasks/main.yml
---
- name: Import Elastic GPG key
  rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present

- name: Add Elastic 8.x repository
  yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: true

- name: Install Logstash
  dnf:
    name: logstash
    state: present

- name: Enable and start Logstash
  systemd:
    name: logstash
    enabled: true
    state: started
```

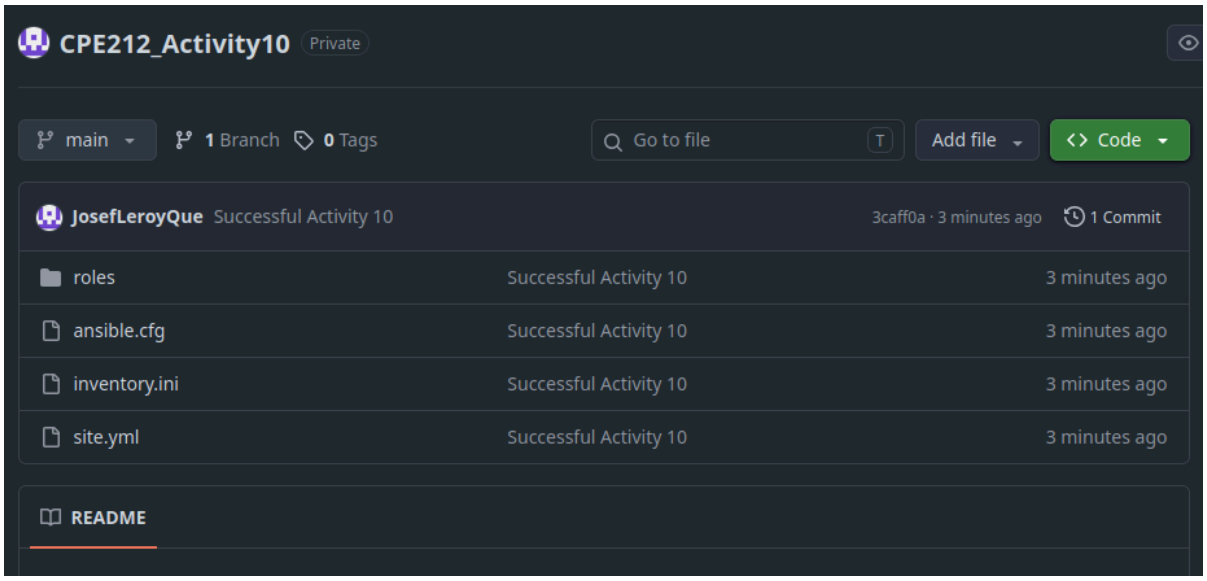
Analysis: This task installs Logstash onto a CentOS machine by utilizing the Elastic GPG key and creating a Yum repository. The Logstash package is then installed using the dnf installer module. Finally, the logstash.service is started via systemd so that it can keep running in the background.

```
leroy1@leroyque:~/CPE212_Activity10$ ssh -t sysadleroy1@192.168.56.107 "sudo systemctl status logstash"
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
[sudo] password for sysadleroy1:
● logstash.service - logstash
     Loaded: loaded (/usr/lib/systemd/system/logstash.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-09-30 15:32:44 PST; 9s ago
   Main PID: 44593 (java)
      Tasks: 16 (limit: 10496)
     Memory: 254.2M (peak: 254.6M)
        CPU: 19.134s
     CGroup: /system.slice/logstash.service
             └─44593 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -Djava.awt.headless=true -Dfile.encoding=UTF-8 -XX:+HeapDumpOnOutO

Sep 30 15:32:44 localhost.localdomain systemd[1]: Started logstash.
Sep 30 15:32:44 localhost.localdomain logstash[44593]: Using bundled JDK: /usr/share/logstash/jdk
lines 1-12/12 (END)
```

Analysis: This checking via SSH shows that Logstash is currently active and working on the CentOS VM.

Git Commit:



Reflections:

Answer the following:

1. What are the benefits of having a log monitoring tool?

- Log monitoring tools are used to collect logs of various servers in one platform to give administrators early visibility into problems and resolve issues faster because they do not have to connect to each server using SSH to check status. Such an approach will allow operations to decrease application downtime and improve security.

2. CONCLUSION

For this laboratory activity, I used Ansible as my Infrastructure as code (IaC) tool for automated deployment of the Elastic Stack on a number of virtual machines with Linux operating systems such as Ubuntu and CentOS. By creating modular roles, I learned how to utilize several Linux package managers and configure different services such as Elasticsearch, Kibana, and Logstash without spending time doing repetitive installation manually. I also learned how to resolve problems with network binding settings, SSH terminal permissions, and service dependencies on managed nodes.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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